

Persistencia Usada:

Se ha utilizado un XML para la persistencia. Al empezar la ejecución busca el archivo XML con el nombre de "Catalogos.xml". Hay una clase específica para el XML (XMLcatalog) donde se crean 5 listas donde apuntan a clases de clave valor (Entry). En algunos casos como en ProductMeeting, donde se encuentra una clase con fecha, hemos utilizado un adapter, que cuando se haga uso de fechas, utilizar ese adapter que transforme la fecha a un String o viceversa.

Gestión de excepciones:

La excepción usada ha sido "IllegalArgumentException" a lo largo del código, que funciona gracias a un try-catch implementado en la clase CLI que evita que la aplicación se detenga de golpe y manda un mensaje informativo.

Ejecución de comandos:

```
tUpm> help
help
Commands:
  client add "<nombre>" (<DNI>|<NIF>) <email> <cashId>
  client remove <DNI>
  client list
  cash add [<id>] "<nombre>"<email>
  cash remove <id>
  cash list
  cash tickets <id>
  ticket new [<id>] <cashId> <userId> -[c|p|s] (default -p option)
  ticket add <ticketId><cashId> <prodId> <amount> [--p<txt> --p<txt>]
  ticket remove <ticketId><cashId> <prodId>
  ticket print <ticketId> <cashId>
  ticket list
  prod add [<id>] "<name>" <category> <price> [<maxPers>]] || ("<name>" <category> )
  prod update <id> NAME|CATEGORY|PRICE <value>
  prod addFood [<id>] "<name>" <price> <expiration:yyyy-MM-dd> <max_people>
  prod addMeeting [<id>] "<name>" <price> <expiration:yyyy-MM-dd> <max_people>
  prod list
  prod remove <id>
  help
  echo "<text>"
  exit
```

```
tUpm> echo "Agrego Libro"
echo "Agrego Libro"
"Agrego Libro"
```

```
tUpm> prod add 1 "Libro P00" BOOK 25
prod add 1 "Libro P00" BOOK 25
{class:Product, id:1, name:'Libro P00', category:BOOK, price:25.0}
prod add: ok
```

```
tUpm> prod addMeeting 23456 "Reunion Rotonda" 12 2026-12-21 100
{class:Meeting, id:23456, name:'Reunion Rotonda', price:12.0, date of Event:2026-12-21, max people allowed:100}
prod addMeeting: ok
```

```
tUpm> prod addMeeting 23457 "Graduacion ETSISI" 40 2026-12-21 30
{class:Meeting, id:23457, name:'Graduacion ETSISI', price:40.0, date of Event:2026-12-21, max people allowed:30}
prod addMeeting: ok
```

```
tUpm> prod list
prod list
-----Productos-----
Catalog:
  {class:Product, id:1, name:'Libro P00', category:BOOK, price:25.0}
  {class:Product, id:2, name:'Camiseta talla:M UPM', category:CLOTHES, price:15.0}
-----
-----Servicios-----
-----
prod list: ok
```

```
tUpm> prod update 1 NAME "Libro P00 V2"
prod update 1 NAME "Libro P00 V2"
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:25.0}
prod update: ok
```

```
tUpm> prod remove 3
prod remove 3
{class:Product, id:3, name:'Libro P00 repetido Error', category:BOOK, price:25.0}
prod remove: ok
```

```
tUpm> cash add UW1234567 "pepecurro3" pepe0@upm.es
cash add UW1234567 "pepecurro3" pepe0@upm.es
Cash{identifier='UW1234567', name='pepecurro3', email='pepe0@upm.es'}

cash add: ok
```

```
tUpm> client add "Pepe3" 55630667S pepe1@upm.es UW1234567
client add "Pepe3" 55630667S pepe1@upm.es UW1234567
USER{identifier='55630667S', email='pepe1@upm.es', name='Pepe3', registeredBy='UW1234567'}
client add: ok
```

```
tUpm> client list
client list
Client:
  USER{identifier='55630667S', email='pepe1@upm.es', name='Pepe3', registeredBy='UW1234567'}
  USER{identifier='98948334B', email='pepe2@upm.es', name='Pepe2', registeredBy='UW1234567'}
  USER{identifier='Y8682724P', email='pepe3@upm.es', name='Pepe1', registeredBy='UW1234567'}
client list: ok
```

```
tUpm> client remove Y8682724P
client remove Y8682724P
client remove: ok
```

```
tUpm> cash list
cash list
Cash:
  Cash{identifier='UW1234567', name='pepecurro3', email='pepe0@upm.es'}

  Cash{identifier='UW1234569', name='pepecurro1', email='pepe0@upm.es'}

  Cash{identifier='UW7883307', name='pepecurro2', email='pepe0@upm.es'}

cash list: ok
```

```
tUpm> cash remove UW1234569
cash remove UW1234569
cash remove: ok
```

```
tUpm> cash tickets UW1234567
cash tickets UW1234567
Tickets:
  No tickets found for this cashier
cash tickets: ok
```

```
tUpm> ticket new UW1234567 55630667S
ticket new UW1234567 55630667S
Ticket : 1 (Products)
  Status: EMPTY
  Client type: Individual
ticket new: ok
```

```
tUpm> ticket new 212121 UW1234567 55630667S
ticket new 212121 UW1234567 55630667S
Ticket : 212121 (Products)
  Status: EMPTY
  Client type: Individual
ticket new: ok
```

```
tUpm> ticket add 212121 UW1234567 1 20
Ticket : 212121
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
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{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
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{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:B00K, price:30.0} **discount -3.0
Total price: 600.0
Total discount: 60.0
Final Price: 540.0
```

```
tUpm> ticket list
```

```
Ticket List:
```

```
UW1234567 1 - EMPTY
```

```
UW1234567 2 - EMPTY
```

```
UW1234567 212121 - CLOSED
```

```
UW1234567 212123 - CLOSED
```

```
UW1234567 212127 - CLOSED
```

```
UW1234567 212128 - CLOSED
```

```
UW1234567 212129 - CLOSED
```

```
UW1234567 212130 - CLOSED
```

```
UW1234567 3 - EMPTY
```

```
ticket list: ok
```

```
tUpm> ticket remove 212123 UW1234567 2
```

```
Ticket : 212123
```

```
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
```

```
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
```

```
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
```

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```

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```

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```

```
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
```

```
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
```

```
Total price: 600.0
```

```
Total discount: 60.0
```

```
Final Price: 540.0
```

```
ticket remove: ok
```

```
tUpm> ticket print 212123 UW1234567
Ticket : 212123-2026-01-18-13:30
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:1, name:'Libro P00 V2', category:BOOK, price:30.0} **discount -3.0
{class:Product, id:2, name:'Camiseta talla:M UPM', category:CLOTHES, price:15.0} **discount -1.1
{class:Product, id:2, name:'Camiseta talla:M UPM', category:CLOTHES, price:15.0} **discount -1.1
{class:Product, id:2, name:'Camiseta talla:M UPM', category:CLOTHES, price:15.0} **discount -1.1
Total price: 645.0
Total discount: 63.1
Final Price: 581.9
ticket print: ok
```

```
tUpm> exit
Closing application.
Goodbye!
```